

Namana Y Tarikere

Arlington, VA | namanaytarikere@gmail.com | (703) 469-8111 | LinkedIn: [/in/namanaty](https://www.linkedin.com/in/namanaty) | GitHub: [/namanaty](https://github.com/namanaty)

Professional Software Developer with 3 years of work experience building and securing enterprise applications, mitigating vulnerabilities and securing cloud platforms. I am skilled in threat analysis, risk assessment and compliance standards like NIST, ISO and OWASP. Passionate about staying updated of evolving technology.

EDUCATION

George Washington University, Washington DC, USA

Aug 2023 – May 2025

Masters in Cybersecurity, Recipient of SEAS Merit Award (GPA: 3.78/4)

Coursework: Computer Security, Network Security, Management of Information Systems, E-Commerce Security, Unix System Administration, Information Policy, Cloud Computing

BMS College of Engineering, Bengaluru, India

Aug 2016 – Sep 2020

Bachelor of Engineering in Information Science (GPA: 3.48/4)

Coursework: Algorithms, Data Structures, Operating Systems, Computer Networks, Information Security, Database Management Systems, Computer Architecture, Computer Organization, Software Engineering

TECHNICAL SKILLS

Languages: Python, Java, C#, C, C++, PHP, Shell Scripting

Frameworks: JavaScript (React.js, Node.js, Angular.js), Microsoft .NET

Cloud Platforms: Pega, AWS, Google Cloud Platform (GCP)

Tools: Docker, Nmap, Splunk, Wireshark, John the Ripper, Burp Suite, Cisco Packet Tracer, Metasploit

Cybersecurity: Load Balancing, STP, VLAN, Zero Trust, MITRE ATT&CK, NIST, SIEM, PKI, SASE

Databases and Operating Systems: SQL, MongoDB, Windows, Mac OS, Ubuntu, Kali Linux, Core

Courses: AWS Academy Cloud Architecting and Developing, AWS Cloud Security Foundations and Builder

Certifications: Pega Certified System Architect version 8.5, Pega Certified Senior System Architect version 8.7, CompTIA Security+ Certification (In Progress)

PROFESSIONAL EXPERIENCE

Software Engineer

Oct 2020 – Aug 2023

LTI Mindtree

Bangalore, India

- Alleviated Citibank's dispute resolution time by 50% by integrating multiple applications into the Personal Investment Landing and Point of Sale (PILPOS) system using Pega PRPC and API integrations to streamline case management
- Decreased operational effort by 60% through optimization of risk assessment of failed transactions and service disputes
- Automated the management of 12 types of cash lending and security devices across all Citibank branches in the U.S.
- Enhanced the efficiency of Virtual Branching System (VBS) by automating ATM maintenance requests with job schedulers and implementing RBAC (Role Based Access Control) for granular access control
- Minimized VBS system downtime using Root Cause Analysis (RCA) tools to resolve critical deployment issues
- Migrated Citibank's RSM UI to Pega Cosmos 8.6 to cut the load time by half and boost UI responsiveness
- Built CI/CD pipelines for seamless application deployment and integration
- Collaborated with compliance teams to ensure systems met data privacy policies and audit framework requirements
- Earned 2 LTI-Excellence awards for technical contributions to Citibank towards risk remediation and system reliability

Technical Intern

Jan 2020 – Jul 2020

Rakuten

Bangalore, India

- Designed a Function as a Service (FaaS) application on Openwhisk platform using React.js, Node.js and MongoDB to incorporate session expiry and web action packaging to optimize resources and improve client service execution by 30%
- Reduced REST API response time by 40% using caching and reduced authentication overhead with RBAC
- Spearheaded the database migration from MongoDB to CouchDB to improve fault tolerance, offline synchronization and data consistency in distributed environments
- Decreased support ticket count of FaaS by 25% with unit tests and bug fixes using the Mocha-Chai-Sinon framework
- Collaborated with cross-functional teams to improve platform reliability and scalability across multiple cloud services

PROJECTS

E-Commerce Security and AWS Infrastructure Hardening

Jan 2025 – Apr 2025

- Deployed a secure AWS-based e-commerce platform using EC2, WordPress, RDS, and Cloudflare SSL/TLS, with Multi-factor authentication and secure SSH login for admin access
- Configured IAM roles, S3 bucket policies, KMS encryption and Secrets Manager to enforce principal of least privilege
- Implemented CloudWatch, CloudTrail for continuous monitoring and logging, SNS for real-time security alerts and automated EC2/EBS backups and RDS snapshots for disaster recovery
- Performed OWASP-based vulnerability assessments and penetration testing, delivering a comprehensive report with remediation strategies to strengthen website security

Network Security: Traffic Analysis and Attack Simulations

Aug 2024 – Dec 2024

- Analyzed IPSec protocols, firewall rules and Wi-Fi security (WPA/WEP) on Wireshark to detect and mitigate vulnerabilities
- Simulated network attacks like SYN flooding, ARP poisoning, and DoS/DDoS to monitor anomalies in traffic patterns
- Evaluated SSL/TLS configurations, firewall efficiency and Intrusion Detection and Prevention systems (IDS/IPS) to prevent unauthorized access and improve network visibility through packet analysis
- Evaluated VPN tunneling protocols and load balancing setups for secure and efficient network traffic management

Ethical Hacking and System Exploitation

Jan 2024 – May 2024

- Analyzed strengths and weaknesses of cryptographic encryption techniques like Vigenère Cipher and Caesar Cipher to demonstrate secure and insecure data handling practices
- Discovered critical vulnerabilities through system exploitation using Nmap, John the Ripper, Burp Suite and Metasploit
- Gained root access on 12 Kali Linux virtual machines using ethical hacking methods like brute-force and social engineering
- Conducted packet sniffing and spoofing exercises to understand attack vectors and exploit protocol-level vulnerabilities

Eye Blink Detection for Password Authentication

Jan 2020 – May 2020

- Implemented a multi-layer authentication system using Haar cascade classifiers, facial recognition and Machine Learning in Python to enhance login security
- Crafted a blink-activated virtual keyboard with cursor-based control to securely input OTPs and mitigate phishing, shoulder surfing and thermal tracking attacks
- Incorporated real-time eye movement tracking to detect blinks to ensure accurate and accessible user authentication
- Developed state-based authentication flow triggered by the real-time video processing logic for facial and eye tracking

Automated Vehicle Entry System

Sept 2019 – Dec 2019

- Developed an IoT-based automated vehicle access system using OpenCV library in Python for license plate detection and PyTesseract OCR engine for character recognition
- Integrated Raspberry Pi, camera and ultrasonic sensors to auto-capture images and verify as registered vehicles approach
- Connected the system to a real-time database to validate the license plates against registered vehicles for secure access
- Optimized the system for accuracy across varying environmental conditions like lighting and weather

PROFESSIONAL AND LEADERSHIP ACTIVITIES

- **Published a comprehensive paper on Review of Machine Learning Algorithms in Healthcare** by researching the applications of machine learning algorithms to analyze diagnostic imaging for predicting early onset of Diabetes, Small Cell Lung Cancer (SCLC) and Alzheimer's diseases to support timely medical intervention
- **Volunteer at International Monetary Fund (IMF):** Supported event coordination and guest check-in at the IMF 2025 Spring Meetings to ensure smooth panel discussions for highly influential global financial leaders and policymakers
- **International Student Community Ambassador at GWU:** Mentored over 120 international students by providing personalized career guidance, resume and cover letter reviews and networking strategies. Organized social, cultural and professional events specifically tailored to support academic and career development of international students
- **Member of Women in Cybersecurity (WiCyS) at GWU:** Organized and participated in three cybersecurity conferences, one ethical hacking workshop, a Capture the Flag (CTF) competition and several active events to promote cybersecurity awareness, education and engagement within the GWU cybersecurity community
- **IEEE Student Member:** Volunteered at IEEE conferences, Annual General Meetings and IEEE Bangalore sector events